An Independent Evaluation of Surya's Solar Flare Forecasting: Cheap Baselines Match a 366M-Parameter Foundation Model

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Abstract

Surya is a 366M-parameter heliophysics foundation model released by NASA and IBM, pretrained on Solar Dynamics Observatory (SDO) imagery. Its accompanying paper reports a solar flare forecasting result of TSS 0.436 (HSS 0.522, F1 0.561) against two deep-learning image baselines. We present, to our knowledge, the first independent evaluation of the released `solar_flares_surya` checkpoint. Streaming raw SDO inputs from the official archive, we scored the model on 1,146 stratified forecast hours spanning 2011–2024 (218 positive), and separately evaluated cheap non-imagery baselines on the *complete* official splits (3,672 validation and 43,848 test hours).

We report five findings, ordered by the strength of their support. (i) **The benchmark's effective sample is far smaller than it appears.** Forecast hours inside a 24-hour block are strongly autocorrelated, so the unit of evidence is the block, not the hour: our 739 validation hours occupy 50 blocks of which only **6 contain any positive**, and 407 test hours occupy 28 blocks of which 11 do. Block bootstrap accordingly yields 95% intervals 0.46–0.81 TSS wide; paired differences straddle zero for eight of ten method pairs, and the strongest separation is the shipped 0.5 threshold scoring significantly below 24-hour persistence on the test window (ΔTSS -0.445, 95% CI $[-0.825, -0.086]$); no pair survives a ten-way family-wise correction, which is the under-powering restated. (ii) **Calibration is regime-dependent and does not transfer.** Surya's probabilities are close to faithful on the validation window (Brier 0.057) and collapse on the test window (Brier 0.208): across 15 independent blocks, hours assigned 0.05–0.25 are followed by a flare 56.6% of the time against a mean predicted 0.148, and a Platt rescaling fitted on validation returns a near-identity map that leaves decision quality unchanged. (iii) **A cheap non-imagery baseline matches or exceeds the model.** An 11-feature logistic regression over the past week of GOES X-ray flux —seconds to train on a CPU—ties Surya on identical validation hours (TSS 0.685 vs 0.673) and exceeds it on identical test hours (0.738 vs 0.632), a paired difference that excludes zero, though only marginally; on the complete splits it reaches 0.661 and 0.554 against the reported 0.436, a comparison we report but flag as unmatched. (iv) **No trivial baseline is reported at all**: 24-hour persistence alone reaches 0.430 on the full validation split, with a 95% interval of $[0.238, 0.621]$ that contains the reported 0.436. (v) **Skill appears regime-split, but only its structural half is estimable.** Persistence cannot anticipate a flare episode's onset by construction, and false-alarms through every decay hour; the model's onset behaviour, however, rests on just 4 and 3 independent onset-containing blocks and therefore supports description but not estimation—we report it as an observation, not a rate.

Findings (ii) and (v), together with the threshold behaviour in Section 4.6, share one mechanism: base-rate non-stationarity. The positive rate in the official test split rises from 0.0055 in 2020 to 0.697 in 2024, a 128-fold shift across the solar cycle. Under a frozen threshold the two cheap methods move in opposite directions, yet the pooled test TSS exceeds every per-year value for both, a textbook Simpson's paradox. Finding (i) has a separate cause, the combination of temporal autocorrelation with event rarity, and finding (iii) a third, the information already carried by the X-ray record.

Independently of our measurements, we document that the reported 0.436 cannot be located: the released artifacts give three mutually incompatible split definitions and the results table names none of them, while the two companion papers report ResNet50 at TSS 0.018 and 0.261 for the same task, a reference level less stable than the margin being claimed.

We therefore argue that the deficiency is not specific to Surya but structural to the evaluation protocol, and we propose concrete fixes: mandatory cheap baselines, per-regime rather than pooled reporting, explicit threshold procedures, and uncertainty intervals computed over temporal blocks.

1 Introduction

Foundation models have begun to arrive in the physical sciences, and heliophysics is among the newest arrivals. Surya, released in August 2025 by NASA and IBM (Roy et al., 2025), is a 366M-parameter transformer pretrained on Solar Dynamics Observatory imagery and published openly with adapted heads for several downstream tasks. Its release was accompanied by SuryaBench (Roy et al., 2026), a benchmark suite whose solar flare forecasting task asks a deceptively simple question: given full-disk solar imagery at time t, will the peak GOES soft X-ray flux exceed M1.0 during the next 24 hours?

Benchmarks of this kind do more than measure models; they define what the field will treat as progress. A reported score becomes the number that later work must beat, the number that funding proposals cite, and the number that operational users read as an indication of readiness. That gives the *design* of the evaluation—which baselines appear, how the score is aggregated, whether uncertainty is quoted—at least as much influence on the field as the model itself.

Independent evaluation of such models is nevertheless rare, for a mundane reason: it is expensive. Each forecast hour in this task requires two SDO timesteps of 13 channels at 4096^2 resolution; scoring even a modest sample means streaming hundreds of gigabytes through a GPU. The practical consequence is that reported numbers tend to go unchallenged, not because they are beyond challenge but because checking them costs more than most groups will spend.

This paper reports what we found when we paid that cost. We evaluated the released `solar_flares_surya` checkpoint with the authors' own inference code on 1,146 stratified forecast hours spanning 2011–2024, and—because they require no GPU at all—we evaluated cheap non-imagery baselines on the complete official splits, 3,672 validation and 43,848 test hours. To our knowledge, no independent evaluation of the released checkpoint has been reported.

Our contributions are six. The first five are measurements, and they compound; the sixth is documentary and independent of anything we ran.

1. **The benchmark reports no cheap baseline.** Its comparators are two image models (AlexNet, ResNet50). A 24-hour persistence rule, which requires no model at all, already reaches TSS 0.430 on the full validation split, with a day-blocked 95% interval of $[0.238, 0.621]$ that contains the 0.436 reported for the foundation model.

2. **A cheap non-imagery model matches or exceeds the foundation model.** On the *same* forecast hours, an 11-feature logistic regression over the past week of GOES X-ray flux

ties Surya on validation (0.685 vs 0.673) and beats it on test (0.738 vs 0.632). On the complete splits it reaches 0.661 and 0.554, above the reported 0.436, a comparison we report but flag as unmatched, since full-split inference of the model was beyond our compute budget.

3. **Single-number comparisons on this task are under-powered.** Forecast hours within a 24-hour block are strongly autocorrelated; resampling whole blocks yields 95% intervals 0.46–0.81 TSS wide; paired block-bootstrap differences straddle zero for eight of ten method pairs, and even the strongest separation—the shipped threshold falling below persistence on test—does not survive a ten-way family-wise correction.

4. **Skill appears regime-split, and the two regimes are complementary.** Persistence is structurally blind to episode onsets and false-alarms on every decay hour, a definitional statement that holds at any sample size. The model's complementary behaviour on onsets is visible in our sample but, as Section 4.4 shows, rests on too few independent episodes to be quoted as a rate.

5. **Calibration is regime-dependent and does not transfer.** Surya's probabilities are close to faithful in the validation window and collapse in the test window; a rescaling fitted on the former transfers as a near-identity map and changes nothing.

6. **The reported score cannot be located in the released artifacts.** Three incompatible split definitions are on offer and the results table names none of them; the two companion papers disagree by a factor of fourteen on the same baseline's TSS; and the dataset documentation states an event threshold one order of magnitude away from the one its own labels follow. This is a reading of the public record, not a measurement, and it is set out with sources in Section 2.3.

Contributions 4 and 5, and the threshold behaviour documented in Section 4.6, share one cause. The positive rate in the official test split rises from 0.0055 in 2020 to 0.697 in 2024 as the solar cycle climbs toward maximum, a 128-fold shift within a single split. Under a frozen decision threshold the cheap methods move in opposite directions, and pooling across the split produces, for each of them, a score higher than that of any individual year. Contribution 3 has a different origin—autocorrelation and event rarity, and contribution 2 a third, namely how much of the 24-hour flare signal the X-ray record already carries.

We therefore frame this work as a critique of an evaluation protocol rather than of a model. The authors themselves describe the downstream heads as "proof-of-concept studies ... not optimized for end-to-end or operational forecasting use," and we take that qualification at face value. Our claim is narrower: *even as a proof of concept, this protocol cannot rank methods*, because the contribution of the imagery modality is never isolated from what X-ray history alone provides. We close with six concrete changes that would restore that ability, all of them cheap.

2 Background

2.1 Surya and SuryaBench

Surya is a 366M-parameter transformer over 13 SDO channels—8 AIA (seven EUV plus one UV at 1600 Å) and 5 HMI (line-of-sight magnetic field, three vector components, Doppler velocity)—at 4096^2 native resolution, released on Hugging Face under Apache 2.0 with LoRA-adapted downstream heads.

We are deliberately careful about the size of the pretraining corpus, because the released artifacts do not agree. The model paper states that the ML-ready SDO database spans "May 13, 2010, to December 31, 2024" and that "the total size of the data for dataset is around 257 TB" (its Section 2.1.2), and separately that the pretraining partition used "observations from 2011 to 2019." The Hugging Face model card instead describes the model as "pretrained on 9 years (≈218 TB) of multi-instrument data." The strings "218" and "nine years" do not appear in the paper. No released source states how the two relate: the paper never gives 218 TB, the model card never gives 257 TB, and neither connects a nine-year pretraining corpus to the full 2010–2024 database. We cite the 218 TB to the model card, its only source, and leave the two figures unreconciled rather than reconciling them on the authors' behalf. This is a small illustration of a pattern documented in Section 2.3.

2.2 The flare forecasting task

Binary: does the peak GOES soft X-ray flux exceed $\theta_{\max} = 10^{-5}\,\mathrm{W\,m^{-2}}$ (M1.0) in the window $[t, t + 24\mathrm{h})$? Hourly cadence. Model input is two timesteps ($t - 60\,\mathrm{min}$ and t). The official splits, as they appear in the released CSVs, are given in Table 1.

Table 1: The official splits as they exist in the released files, counted directly. The dataset card describes validation and `leaky_validation` as covering 2010–2019, but 2010 is absent from both (3,672 = 17 days × 24 h × 9 years).

split	years	calendar window	hours	base rate
train	2010–2019	Feb 15 – Dec 31	74,760	0.1211
validation	2011–2019	Jan 15–31	3,672	0.1089
`leaky_validation`	2011–2019	Jan 1–14, Feb 1–14	6,048	0.1490
test	2020–2024	full year	43,848	0.2943

Two features of this design matter later. The validation split is a set of seventeen-day January windows drawn from the *same* years as training, which the authors implicitly acknowledge by naming the adjacent split `leaky_validation`. And the test split spans the rising phase of solar cycle 25, so its base rate is not stationary (Section 4.6).

2.3 Reported results and why they cannot be located

Table 4 of the model paper reports Surya at TSS 0.436 / HSS 0.522 / F1 0.561 against AlexNet (0.358 / 0.398 / 0.454) and ResNet50 (0.018 / 0.028 / 0.055). Both comparators are image models. **The evaluation split is not stated; the decision-threshold procedure is not stated; no uncertainty is reported; no non-imagery baseline appears.** We read that section in full: it defines the label, then the metrics, then presents the table. There is no train/validation/test partition for this task, no date range, no sample count, and no description of how a probability is converted into a binary forecast.

This is worse than an omission, because the released artifacts offer three *mutually incompatible* candidates for the missing split (Table 2).

The model paper's partition is given in the SDO-data section and never linked to the downstream tasks; the SuryaBench figure is for the core dataset, with file counts (379,920/43,680/43,800) that do not correspond to the hourly flare task at all. Since the results table names none of them, **TSS 0.436 is not attributable to any specific evaluation period.** Our own reading—that it most plausibly refers to the Jan 15–31 validation window—is an inference, and we mark it as one wherever it matters.

Table 2: Three incompatible candidate splits, none of which the results table names.

source	training	validation	test
model paper, SDO-data section	2011–2019, days 46–365	—	2011–2019, days 15–31
SuryaBench, core SDO data	2010–2018	2019	2020
Hugging Face flare dataset card	2010–2019, Feb 15–Dec 31	Jan 15–31	2020–2024

A second discrepancy is sharper still. The two companion papers, submitted the same day by overlapping author lists, report irreconcilable numbers for the same baseline architectures on the same task:

Table 3: The same baselines, the same task, two companion papers.

model	Surya paper, Table 4	SuryaBench, Table 3(b)
AlexNet	TSS 0.358, HSS 0.398, F1 0.454	TSS 0.359, HSS 0.354, F1 0.679
ResNet50	TSS **0.018**, HSS 0.028, F1 0.055	TSS **0.261**, HSS 0.281, F1 0.627

In Table 3, ResNet50's TSS differs by a factor of about fourteen and AlexNet's F1 by 0.225. Which is right we cannot say, and that is what matters here: the reference level against which the foundation model's 0.436 is judged is itself unstable by more than the margin being claimed. Whatever explains the gap—a different split, a different threshold, a different training run—is exactly the information neither table records.

We note one further inconsistency because it affects anyone reproducing the task. The SuryaBench text says its binary thresholds correspond "to the equivalent strength of an M1.0-class flare" while stating the threshold as $10^{-4}\,\mathrm{W\,m^{-2}}$, which is X1.0, not M1.0. The model paper uses $10^{-5}\,\mathrm{W\,m^{-2}}$ (M1.0). We resolved the ambiguity empirically against the released labels themselves: they follow M1.0, as verified on all 128,328 rows in Section 2.4.

2.4 A label-leakage trap in the released data

In the released flare CSVs, `max_goes_class[t]` is the maximum class over $[t, t+24\mathrm{h})$—i.e. it is the source of the label, not a past observation. We verified `label_max` == (`max_goes_class` $\geq$ M1.0) on 128,328 of 128,328 rows. Any baseline that reads this column at time t is therefore trivially perfect and meaningless. The same holds for `cumulative_index`: the dataset card defines it over the same prediction window, and we verified the exact identity `label_cum` == (`cumulative_index` ≥ 10) on 128,328 of 128,328 rows. All features in this work are read at $t-24\mathrm{h}$ or earlier; at that lag both forward-looking columns cover $[t-24\mathrm{h}, t)$ and are legitimately observable at forecast time.

2.5 Related work: a standard practice this benchmark omits

Solar flare forecasting already has a community standard for comparative evaluation. Barnes et al. (2016), reporting the 2009 interagency "all-clear" workshop in Boulder, established the methodology and concluded that "there is no single method that is clearly better than the others for flare prediction in general," with no participating method proving substantially better than climatology. The follow-on series—Leka et al. (2019a), Leka et al. (2019b) and Park et al. (2020)—arose from a separate 2017 meeting at the Institute for Space-Earth Environmental Research, Nagoya University, and applied the methodology to operational forecasting systems worldwide. Its central conclusion is directly relevant here. In the words of Paper II's abstract:

> "Numerous methods performed consistently above the 'no skill' level, although which method scored top marks is decisively a function of flare event definition and the metric used; there was no single winner."

That series also normalised reporting climatological reference levels and examining metric sensitivity rather than quoting one number.

We therefore make no claim to have discovered that simple methods are competitive; that result is nearly a decade old, and our Section 4.2 is a re-demonstration of it in a new setting. Our contribution is the observation that a prominently released 2025 foundation-model benchmark **departs from these established practices**: it reports a single pooled TSS against two image models, without a non-imagery or climatological reference, without stating the evaluation split or the threshold procedure, and without uncertainty. Read against Barnes et al. (2016) and the Leka/Park series, the reported comparison cannot support a ranking claim—and our measurements show concretely what that omission conceals: metric-dependent ordering (Section 4.2), interval overlap (Section 4.3), regime-split skill (Section 4.4), and non-transferable calibration (Section 4.5).

2.6 Reference forecasts in research and in operations

Comparison against explicit reference forecasts is not a preference of ours; it is what this field already does, and the operational agency applies it to its own product. NOAA's Space Weather Prediction Center issues daily probabilistic M- and X-class flare forecasts—percentages from 1% to 99% for a specified 24-hour day, at lead times of one to three days, and its published verification measures their skill *against short-term (30-day) climatology and 1-day persistence*, defining negative skill as no improvement over a constant climatological forecast (NOAA SWPC, n.d.). The same expectation runs through the research literature: Leka et al. (2019a) argue that in operational settings the appropriate standard is "the best 'unskilled' forecast available," and the TSS itself—the Hanssen–Kuipers discriminant, or Peirce skill score—was adopted as the standard flare metric precisely to make cross-method comparison meaningful (Bloomfield et al., 2012).

Two recent results make the omission we document harder to excuse. First, in the SHARP-parameter tradition (Bobra et al., 2014) that has dominated machine-learning flare forecasting since Bobra & Couvidat (2015)—who report a TSS of 0.76 in their operational configuration for $\geq$M1.0 within 24 hours—feature-ranking studies repeatedly find prior flare activity among the strongest individual predictors (Nishizuka et al., 2017; Campi et al., 2019), and van der Sande et al. (2023), forecasting M-class flares at 24 hours from magnetogram data, conclude that "flaring history has greater predictive power than our CNN-extracted features." Our Section 4.2 is that finding again, against a much larger image model. Second, and most directly, Camporeale & Berger (2025) verified 27 years of SWPC flare forecasts and report that "even the simple persistence model—using no training and based solely on the previous day's flare activity—performs on par with, or only marginally below, the SWPC forecast." They close with a recommendation we would have been happy to write ourselves:

> "Any solar flare forecasting or all-clear prediction models developed in a research setting should perform the type of basic forecast verification study shown here — using the same baseline models and metrics for comparison — before being claimed as an advance over current methods."

That was published in *Space Weather* in the same year as the model we evaluate. Our contribution is not the idea of checking against persistence; it is that nobody had checked *this* model, and that the benchmark as published does not let a reader do so.

We do not overstate the SHARP literature. We did not find a study that pits a flare-history-only model directly against a SHARP-parameter model at equal footing with both scores reported, and we do not claim one exists; van der Sande et al. (2023) is the closest, and it compares flare history against CNN-extracted magnetogram features rather than SHARP scalars. Nor did we run a SHARP baseline ourselves (Section 7).

3 Methods

3.1 Model evaluation

Released checkpoint `nasa-ibm-ai4science/solar_flares_surya` with the authors' own `infer.py`, `SolarFlareDataset`, `config_infer.yaml` and scalers; raw netCDF inputs streamed from the official S3 archive (download → infer → delete). bf16 autocast on a T4. Deterministic: seed 42, no shuffling, no augmentation. Every label we scored was re-checked against the official split CSVs: 0 mismatches on 739 validation and 407 test hours. Pinned revisions: checkpoint `ec7c42a` (2025-08-18), flare dataset `bf474bc` (2025-12-16; a comparison of the label files shows that revision touched only the dataset README).

3.2 Sampling

Full-split inference is prohibitive. Each forecast hour needs two timesteps at ∼586 MB each, and consecutive hours within a block share one of them; scoring our sample required 1,224 distinct netCDF files, roughly 700 GB streamed. Covering the full test split alone would have required about 25 TB.

We therefore drew a seeded, stratified sample of contiguous 24-hour windows—36 validation windows (2011–2019) and 20 test windows (2020–2024)—deliberately over-sampling flare-active periods so that positives are sufficiently represented. Missing archive files cost 125 of the 864 planned validation hours and 73 of the 480 planned test hours, about 15% in each case, and those gaps fragment some windows: the 36 planned validation windows survive as 50 contiguous runs (20 of at least 20 hours, 30 shorter fragments), and the 20 planned test windows as 28 runs (14 of at least 20 hours). Throughout this paper a *block* means one of these recovered contiguous runs, since that is the unit within which hours are actually autocorrelated.

The resulting sample base rates (0.115 validation, 0.327 test) are higher than the full-split rates; absolute values are therefore not directly comparable to full-split numbers, but every model comparison is made on identical hours.

Sampling sanity check. Because the cheap baselines need no GPU we could score them both on our sample and on the complete splits. Estimates agree closely (GOES logistic 0.685 vs 0.661; persistence 0.405 vs 0.430), indicating that the stratification does not materially distort model ranking.

3.3 Baselines

1. **Climatology**—always predict "no flare".
2. **Persistence**—predict the label observed at $t - 24$h, read from the complete hourly record rather than from the split file, so that no hour is dropped merely because its reference falls in a neighbouring split.

3. **GOES-history logistic regression**—11 features from lags $t-24\text{h} \ldots\ t-168\text{h}$ (log peak flux per lag, 7-day max, cumulative-index terms), standardised, L2-regularised, fitted by gradient descent on the official train split (74,564 of 74,760 rows; 196 dropped for non-finite features, see Section 7). Trains in seconds on a CPU.

3.4 Metrics and uncertainty

TSS (POD − POFD), HSS, F1, full confusion matrices, and a full threshold sweep. Hourly samples within a 24-hour block are strongly autocorrelated, so naive per-sample bootstrap understates variance; we resample **whole blocks** (50 validation, 28 test for the sampled hours; calendar days for the complete splits) and report percentile 95% intervals. Each interval reseeds its own generator, so a printed interval reproduces on its own rather than depending on how many intervals were drawn before it.

4 Results

4.1 The missing baseline

On the full validation split, persistence alone reaches TSS 0.430 with a day-blocked 95% interval of $[\mathbf{0.238}, \mathbf{0.621}]$, an interval that contains the 0.436 reported for a 366M-parameter model. Because no uncertainty is reported alongside the 0.436, the two numbers cannot be formally compared. What the interval does show is that the sampling variability of the *trivial* baseline alone covers the reported result, the information a benchmark table should make visible, and this one does not. On the full test split persistence reaches 0.535 $[0.502, 0.568]$.

4.2 A cheap non-imagery model matches or exceeds the foundation model

Matched comparison (identical hours, our measurements only). This is the primary result because both numbers come from the same forecast hours.

Table 4: Matched comparison on identical forecast hours.

split	hours	GOES-history logistic	Surya (tuned)	Surya @0.5
validation	739	**0.685**	0.673	0.425
test	407	**0.738**	0.632	0.173

On validation the two are indistinguishable; on test the cheap model leads on every metric (HSS 0.699 vs 0.571, F1 0.807 vs 0.735), and the paired block-bootstrap TSS difference excludes zero, though only marginally (+0.106, 95% CI $[+0.009, +0.280]$; Section 4.3). Both thresholds are tuned in-sample, which flatters both equally.

Unmatched comparison (full splits, reported number). Because the baselines need no GPU we could also score them on every official hour, with a no-cherry-picking protocol: weights from the train split, a single threshold chosen on the full validation split (0.10) and frozen for test.

Because Section 2.3 leaves three candidate readings of the reported split, we checked the cheap baseline against all three. Under the model paper's partition (days 15–31 of 2011–2019, i.e. exactly the validation split) it scores 0.661; under the dataset card's (2020–2024) it scores 0.554; under SuryaBench's (2020 alone) it scores 0.452 (Section 4.6). Every one exceeds 0.436, so the comparison

Table 5: Complete official splits under a frozen-threshold protocol. Intervals are day-blocked bootstrap.

split	n	base	model	TSS	95% CI	HSS	F1
validation (full)	3,672	0.109	GOES logistic	**0.661**	$[0.532, 0.777]$	0.375	0.475
validation (full)	3,672	0.109	persistence	0.430	$[0.238, 0.621]$	0.428	0.491
test (full)	43,848	0.294	GOES logistic	**0.554**	$[0.526, 0.581]$	0.436	0.655
test (full)	43,848	0.294	persistence	0.535	$[0.502, 0.568]$	0.536	0.672

does not depend on resolving the ambiguity—and on validation the cheap model's own interval excludes the reported value. **We flag this comparison as unmatched:** the baseline was run on all hours, the model on our sample, and the 0.436 is the authors' own measurement rather than ours. It is offered as corroboration of the matched result above, not as the primary claim. The day-blocked intervals are also robust to the blocking unit: with 2-day and 3-day blocks the validation persistence interval still contains 0.436 and the validation GOES interval still excludes it (`gaps_audit.py`).

The advantage is metric-dependent. Against the *full validation split*, the reported HSS (0.522) and F1 (0.561) exceed the cheap baseline's 0.375 and 0.475, the baseline buys its TSS with a high false-alarm rate (740 false alarms against 355 hits). We therefore do not claim uniform superiority; we claim that on TSS, the base-rate-independent metric standard in this literature, the imagery modality shows no measurable advantage over X-ray history on this task. On the matched test hours the cheap model does lead on all three metrics simultaneously.

4.3 Single-number comparisons are under-powered

Table 6: Block-bootstrap intervals on our measured hours.

split	model	TSS	95% CI	width
validation	Surya @0.5	0.425	$[0.028, 0.746]$	0.72
validation	Surya @0.16 (tuned)	0.673	$[0.289, 0.886]$	0.60
validation	persistence	0.405	$[-0.019, 0.792]$	0.81
validation	GOES logistic @0.10	0.685	$[0.417, 0.881]$	0.46
test	Surya @0.5	0.173	$[0.000, 0.484]$	0.48
test	Surya @0.04 (tuned)	0.632	$[0.315, 0.876]$	0.56
test	persistence	0.618	$[0.258, 0.889]$	0.63
test	GOES logistic @0.34	0.738	$[0.467, 0.940]$	0.47

Intervals are 0.46–0.81 TSS wide. Overlap between individual intervals is, however, weak evidence about a difference; the same resampled blocks move both scores together, so we resample pairs as well: one block draw per iteration scores every method, and the interval of the paired difference is what decides separation (`paired_diff.py`; 4,000 draws; thresholds fixed at their full-sample values).

Eight of ten paired differences straddle zero (the two hybrid pairs, Section 4.4, also straddle). The two that do not are instructive. The cheap baseline's advantage over tuned Surya on the test hours excludes zero only marginally (lower bound +0.009), and we read it as suggestive rather than decisive. The strongest separation in the study is negative: **at its shipped 0.5 threshold**

Table 7: Paired block-bootstrap differences in TSS. Bold rows exclude zero.

split	paired difference	ΔTSS	95% CI
validation	GOES logistic − Surya (tuned)	+0.012	[−0.325, +0.473]
validation	Surya (tuned) − persistence	+0.268	[−0.153, +0.658]
validation	GOES logistic − persistence	+0.279	[−0.058, +0.681]
validation	Surya @0.5 − persistence	+0.019	[−0.398, +0.427]
test	GOES logistic − Surya (tuned)	**+0.106**	**[+0.009, +0.280]**
test	Surya (tuned) − persistence	+0.014	[−0.210, +0.253]
test	GOES logistic − persistence	+0.120	[−0.009, +0.330]
test	Surya @0.5 − persistence	**−0.445**	**[−0.825, −0.086]**

the model scores significantly below 24-hour persistence on the test window (−0.445, [−0.825, −0.086]).

Both exclusions, however, are made at the conventional 95% level while ten comparisons run simultaneously, and neither survives that accounting: the bootstrap tail fractions are 0.55% and 0.70% against the 0.25% a ten-way Bonferroni correction demands, and the corrected 99.5% intervals reach zero ([0.000, +0.382]) or cross it ([−0.924, +0.070]) (`gaps_audit.py`). We therefore describe the shipped-threshold deficit as the strongest separation in the study rather than an unconditional one, and note that even our clearest gap failing a family-wise correction is finding (i) restated: at 50 and 28 blocks, this protocol resolves almost nothing.

The reason differences this large are needed is visible once blocks are counted rather than hours: of 50 validation blocks only **6 contain any positive**, and of 28 test blocks only 11 do. Since the hit-rate term of TSS is determined entirely by positive-containing blocks, the effective sample behind every published flare-forecasting TSS is far smaller than the quoted hour count suggests. **We recommend that the number of positive-containing independent blocks be reported alongside any such score**; it is a one-line addition that would prevent most over-reading.

4.4 Skill appears regime-split, and a trivial hybrid wins

Table 8: Positives partitioned by the persistence reference.

	validation	test
onset positives (persistence blind)	46	27
continuation positives	39	106
decay hours (persistence false-alarms)	35	49
onsets caught — persistence	0/46	0/27
onsets caught — Surya (tuned)	35/46	13/27
onsets caught — Surya @0.5	14/46	0/27
decay false alarms — persistence	35/35	49/49
decay false alarms — Surya (tuned)	10/35	38/49

The two columns differ in kind. The persistence column is not a statistical result: a rule that copies the label from $t - 24$h cannot flag an episode that had not yet begun, and must flag every hour of a decaying one. Those entries are definitional and hold at any sample size.

The model's onset behaviour is a different matter, and its limits are severe. The 46 validation

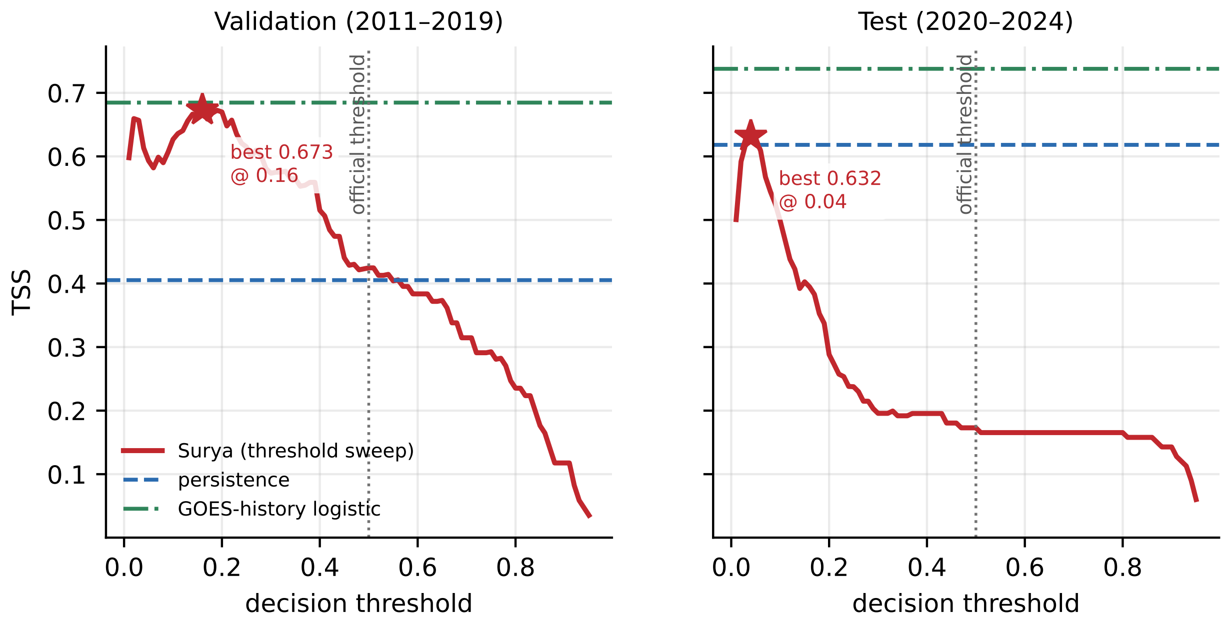

Figure 1: TSS as a function of the decision threshold, with the two cheap baselines as reference lines. The shipped 0.5 threshold sits far from any optimum in both windows.

and 27 test onset hours occupy only **4 and 3 independent blocks** respectively, a handful of flare episodes, not a sample of onsets. Block-bootstrap intervals for the catch rate are correspondingly useless.

Table 9: Onset catch rates. Only the tuned rows carry information, and barely.

split	predictor	caught	rate	95% CI	blocks with a hit
validation	Surya @0.16 (tuned)	35/46	0.761	$[0.357, 1.000]$	4/4
validation	Surya @0.50 (shipped)	14/46	0.304	$[0.000, 0.750]$	3/4
test	Surya @0.04 (tuned)	13/27	0.481	$[0.000, 1.000]$	2/3
test	Surya @0.50 (shipped)	0/27	0.000	$[0.000, 0.000]$	0/3

The last row invites a misreading. A bootstrap interval of $[0.000, 0.000]$ around a 0/27 observation is a boundary artefact of resampling a sample that contains no successes, not evidence of precision. For a zero count the appropriate one-sided statement is the rule of three applied to the *effective* sample: with three onset-containing blocks the 95% upper bound is $3/3 \approx 1.0$, which constrains nothing at all.

We therefore report the onset results as **descriptive observations on a handful of flare episodes, not as rate estimates**: in the three test episodes sampled, the model at its shipped threshold flagged none of the 27 constituent hours, and at a tuned threshold flagged hours in two of the three. The observation is striking and worth checking at scale, but it is not a measured miss rate. The same caution applies to the validation column, where the tuned threshold flags hours in all four episodes.

The two predictors are nonetheless complementary rather than competing, and that statement rests on the pooled score rather than on the onset counts: *persistence OR Surya* reaches TSS 0.705 on validation, above either component (0.673 / 0.405), a point-estimate ordering; the paired difference against the stronger component straddles zero ($+0.032$, $[-0.044, +0.241]$).

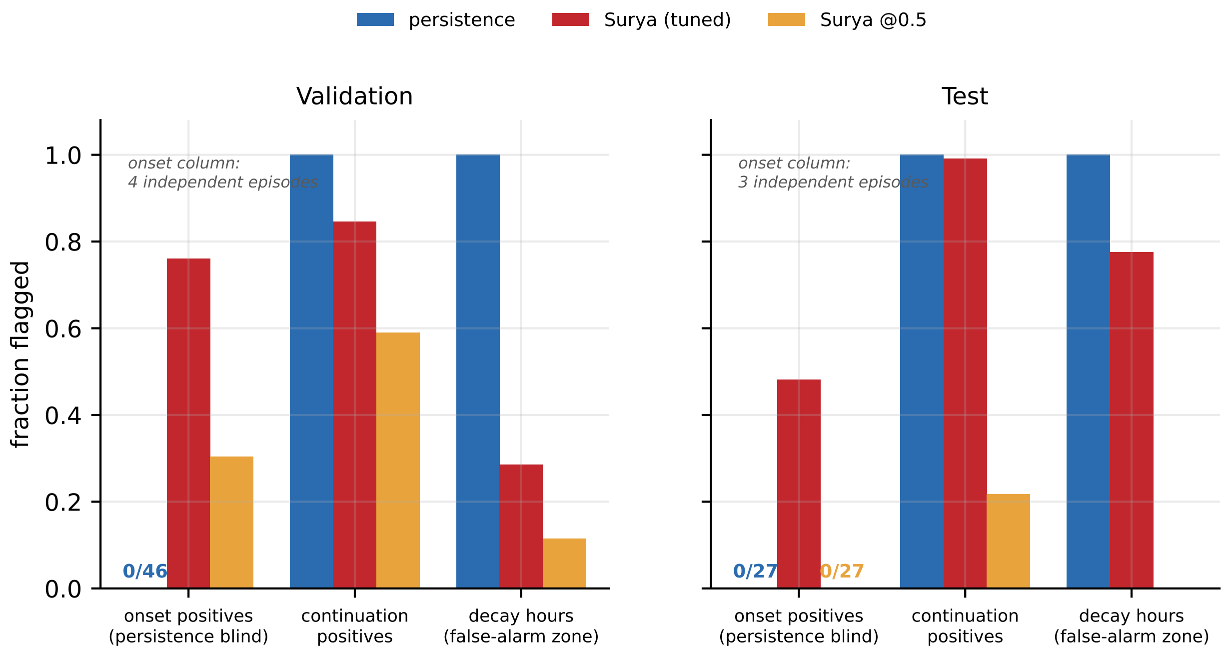

Figure 2: Complementary skill by regime. Persistence is structurally blind to onsets and false-alarms through every decay hour. The onset column rests on four (validation) and three (test) independent episodes and is descriptive only.

4.5 Calibration is regime-dependent and does not transfer

Table 10: Complete reliability tables, with the number of independent blocks contributing to each bin. We show both in full rather than the flattering rows only.

Validation (2011–2019), Brier 0.057					Test (2020–2024), Brier 0.208				
bin	n	blk	pred	obs	bin	n	blk	pred	obs
$[0.00, 0.01)$	391	32	0.002	0.000	$[0.00, 0.01)$	180	15	0.003	0.078
$[0.01, 0.05)$	110	20	0.024	0.118	$[0.01, 0.05)$	41	8	0.021	0.049
$[0.05, 0.10)$	48	15	0.076	0.062	$[0.05, 0.10)$	35	13	0.078	0.657
$[0.10, 0.25)$	88	15	0.172	0.125	$[0.10, 0.25)$	108	13	0.170	0.537
$[0.25, 0.50)$	58	11	0.350	0.362	$[0.25, 0.50)$	20	9	0.307	0.650
$[0.50, 0.75)$	18	4	0.622	0.667	$[0.50, 0.75)$	1	1	0.504	1.000
$[0.75, 1.01)$	26	3	0.866	0.962	$[0.75, 1.01)$	22	1	0.931	1.000

Both tables inherit the stratification of Section 3.2— flare-active windows are over-represented, which lifts the observed frequency in every bin, so the two windows are comparable with each other, but neither is a full-split calibration measurement (Section 7). Validation calibration is good but not perfect: the $[0.01, 0.05)$ bin is under-confident by a factor of about five, in the *same direction* as the test failure. The difference between the two windows is therefore one of degree and extent rather than of kind, but the degree is large, and on test the miscalibration reaches the bins where operational decisions are actually made. Two further caveats belong here: the test $[0.75, 1.01)$ bin, where the model looks excellent, comes from a single block, and the $[0.50, 0.75)$ bin contains one hour.

The combined $[0.05, 0.25)$ band is the most robust statement available: 143 test hours across 15 independent blocks, mean predicted 0.148, observed frequency 0.566. Per-block observed frequencies within that band are strongly bimodal (five blocks at 1.00, five at 0.00), which is itself consistent with an episodic rather than a smoothly-varying error.

A one-parameter Platt rescaling fitted on validation returns $a \approx 0.944$, $b \approx 0.145$—essentially the identity—and leaves test TSS unchanged at 0.173 (Brier improves only from 0.208 to 0.198). The miscalibration is thus not a fixed model bias correctable from in-distribution data; it is a shift tied to the observing regime.

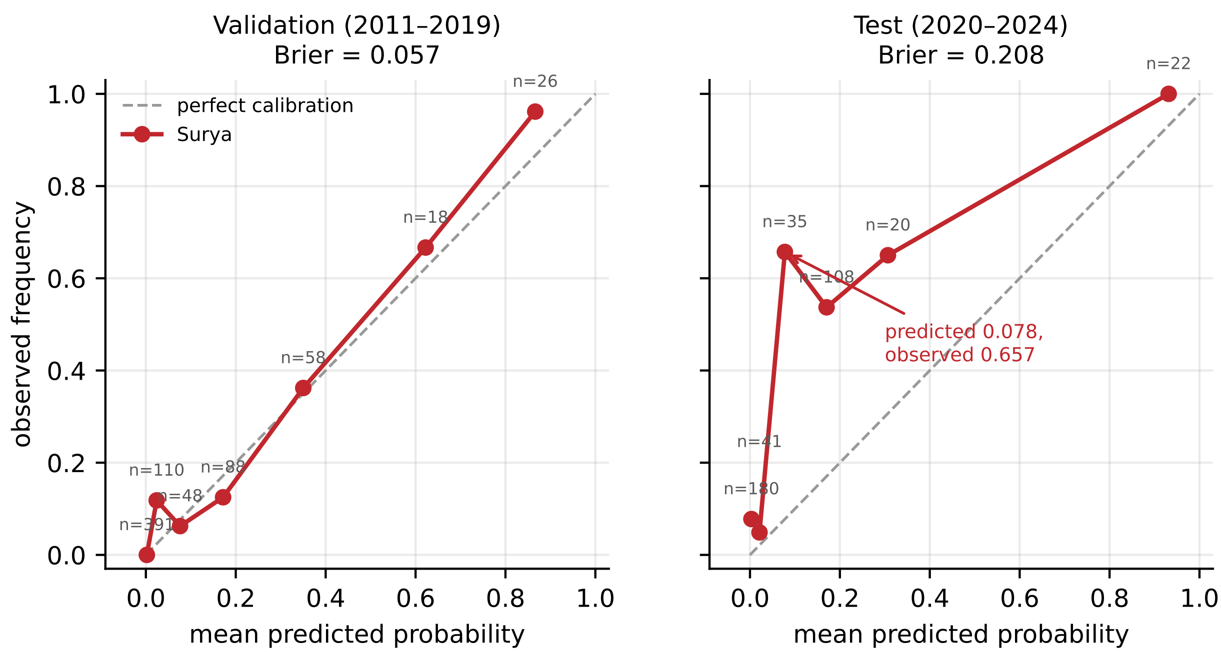

Figure 3: Reliability diagrams. Calibration holds at solar minimum and collapses at maximum; bin counts are annotated because several bins rest on very few independent blocks.

4.6 The common cause: base-rate non-stationarity

Table 11: Per-year test statistics with the validation-frozen threshold.

year	n	base rate	GOES logistic TSS	persistence TSS
2020	8,784	0.0055	0.452	−0.005
2021	8,760	0.0613	0.514	0.270
2022	8,760	0.2638	0.250	0.322
2023	8,760	0.4435	0.056	0.293
2024	8,784	0.6969	0.079	0.389

The positive rate rises 128-fold across the split ($0.0055 \rightarrow 0.6969$), and the two methods respond in opposite directions: the frozen-threshold logistic collapses from 0.514 in 2021 to 0.079 in 2024, while persistence climbs from −0.005 to 0.389. Neither escapes the aggregation artefact, however. Pooled test TSS is 0.554 for the logistic against a per-year range of 0.056–0.514, and 0.535 for persistence against a range of −0.005 to 0.389: for both, pooling manufactures skill that no individual year shows.

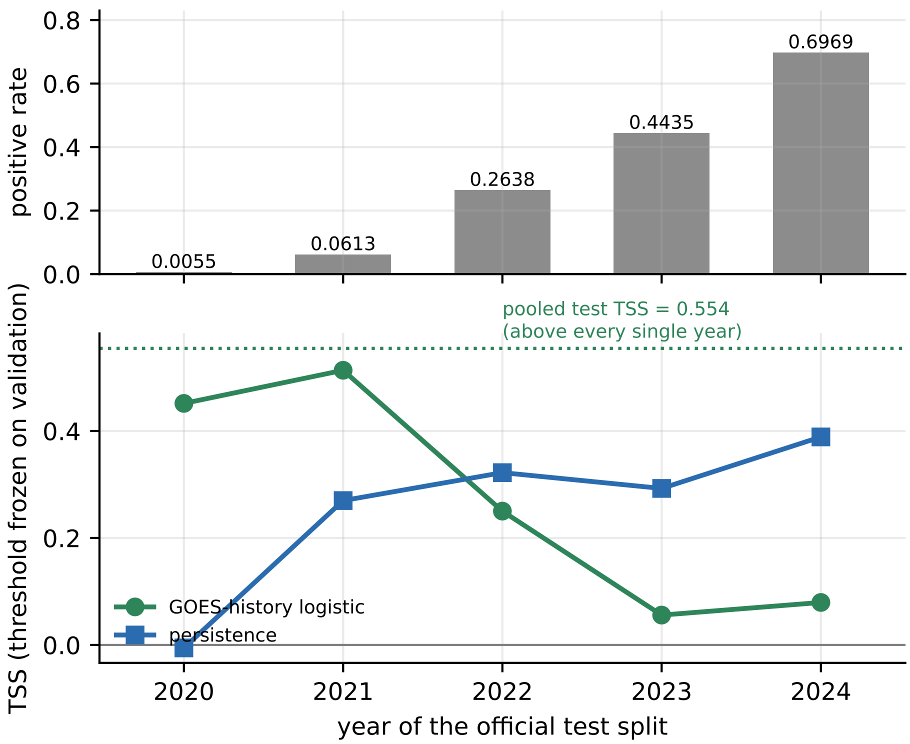

Figure 4: A 128-fold base-rate shift across the official test split breaks every frozen threshold, and the pooled score sits above every individual year.

5 Discussion

5.1 Without a cheap baseline, a modality's contribution is unmeasurable

The question a flare benchmark should answer is not "how well does this model score?" but "what does solar imagery add over what we could already do?" That question is answerable only if the evaluation includes something that does not use imagery. The reported comparison sets Surya against AlexNet and ResNet50—both of which consume the same images, so every entry in the table shares the modality under test, and the marginal value of imagery cancels out of the comparison. Our results suggest that on this particular task the margin is small or absent: a model with eleven coefficients and a single input channel matches or exceeds it.

This should not be read as a verdict on the pretrained backbone. It is a verdict on one binary downstream task, and a plausible reading is simply that 24-hour M-class occurrence is largely determined by whether the Sun is *already* flaring—information fully contained in the X-ray record. If so, the task is a poor instrument for measuring what a solar foundation model knows, and better instruments exist: spatially resolved products, lead-time curves at longer horizons, magnitude rather than occurrence, or transfer across the other downstream tasks released with the model. Demonstrating value where cheap features cannot reach would be a stronger claim than a marginal win where they can.

5.2 Non-stationary base rates make pooled scalar metrics misleading

The solar cycle is the dominant nuisance variable in this problem, and the official test split spans its steep rising phase. A 128-fold change in the positive rate means that the 2020 task and the 2024 task are, statistically, different problems that happen to share a label definition. Pooling them yields a number (0.554) larger than any per-year value (0.056–0.514), a Simpson's paradox produced not by a modelling error but by aggregation across a covariate shift. Reporting that single number invites exactly the wrong inference.

The same mechanism explains the calibration result. A model whose probabilities are faithful when flares are rare is systematically underconfident when they become common, and the failure of Platt rescaling to transfer shows that this is not a fixed bias but a function of the regime. Practically, this means any deployed system must either re-calibrate online or condition explicitly on cycle phase; a calibration certified during solar minimum is not evidence about solar maximum.

We note that this pathology is not specific to heliophysics. Any rare-event forecasting benchmark with a slow cyclic driver—seasonal epidemiology, wildfire risk, certain financial regimes—will produce inflated pooled metrics and non-transferable calibration if evaluated the same way. The fix is not more data but stratified reporting.

5.3 Complementarity is more useful than ranking

Our regime split shows that the two predictors fail in disjoint places. Persistence cannot, even in principle, anticipate an episode that has not begun, and it necessarily false-alarms as an episode decays; those two statements are definitional. In our sample the model covers part of that blind spot, though—as Section 4.4 insists—on too few episodes to quote a rate, and only at thresholds far below the one shipped. What can be defended quantitatively is the pooled result: their disjunction outperforms both components. For an operational user that is the actionable finding—not which single forecaster is better on average, but that a trivial combination covers a structural blind spot. It also suggests that benchmark tables listing models in rank order obscure the more valuable information: *where* each method's skill lives.

5.4 Uncertainty is not optional

With 50 and 28 independent forecast blocks, and only 6 and 11 of them containing any positive—95% intervals span 0.46 to 0.81 TSS. Differences of 0.05–0.10 between methods, the differences that benchmark tables are usually built to display, are therefore not resolvable at this sample size; across our ten paired comparisons even the largest gap, 0.445, fails a family-wise correction (Section 4.3). Reporting point estimates without intervals gives a false impression of resolution, and the standard per-sample bootstrap makes this worse by ignoring within-block autocorrelation and understating variance. Blocked resampling costs nothing and should be routine.

6 Recommendations

1. Report climatology, persistence, and a statistical (non-imagery) baseline alongside every deep model. This is not a new ask: NOAA SWPC verifies its own flare forecasts against 30-day

climatology and 1-day persistence (NOAA SWPC, n.d.), and Camporeale & Berger (2025) recommend exactly this of research models before they are "claimed as an advance over current methods."

2. State the evaluation split, years, and the threshold-selection procedure.
3. Report per-year or per-regime metrics; if pooled numbers are given, show the stratified values beside them.
4. Report uncertainty using temporally blocked resampling, not per-sample bootstrap, and quote the number of positive-containing blocks.
5. Publish reliability diagrams; treat calibration transfer across solar-cycle phases as an explicit evaluation axis.
6. Keep the paper, the companion benchmark paper, the model card and the dataset card consistent with one another and with the released files, and state the event threshold numerically wherever it appears. Every discrepancy in Section 2.3 would have been caught by one pass reconciling the four documents against the data.

7 Limitations

- Model scoring uses a stratified 1,146-hour sample, not the full splits; absolute values are sample-conditioned (mitigated by Section 3.2's sanity check).
- The reliability tables condition on the model's stated probability, but the sampling over-represents flare-active windows, which inflates the observed frequency in every bin relative to the full splits. The validation–test contrast is computed under the same sampling and survives it; absolute statements such as "close to faithful on validation" are sample-conditioned, and the sanity check of Section 3.2 covers TSS, not calibration.
- Both Surya's and the baseline's tuned thresholds in Sections 4.3–4.5 are optimised in-sample, which flatters both; Section 4.2's full-split protocol freezes the threshold on validation instead.
- The full-split baseline numbers are **not matched** to our model measurements; they corroborate but cannot substitute for the matched comparison in Section 4.2.
- Onset-level statements rest on 4 and 3 independent onset-containing blocks. No onset rate—including the zero-catch observation—is estimable at this effective sample size; Section 4.4 reports them descriptively for that reason.
- Even the headline scoreboard rests on 6 (validation) and 11 (test) positive-containing blocks. Our own numbers inherit the limitation we document, which is why every one of them is quoted with an interval.
- Missing archive files cost about 15% of the planned hours (125 of 864 validation, 73 of 480 test) and fragment some 24-hour windows into shorter runs. Because we treat each recovered run as the resampling unit, this fragmentation is reflected in the intervals rather than hidden by them; but it does mean our blocks are not uniformly 24 hours long.

- 196 of 74,760 training rows (0.26%) were dropped for non-finite features. These trace to four hours in the released record whose GOES class is `A0.0`, giving a log-flux of $-\infty$ that propagates into up to seven lagged features each.

- Inference used bf16 autocast. We did not have GPU access for a confirmatory fp32 run, so we bounded instead what one could change (`precision_sensitivity.py`). At the shipped 0.5 threshold **no hour in either split lies within 10^{-3} of the decision boundary**, the closest are 0.0059 (validation) and 0.0039 (test) away, and an adversarial worst case in which every hour within 10^{-2} flips the damaging way moves TSS by at most 0.002 and 0.008 respectively. At the tuned thresholds the same worst case moves TSS by at most 0.033. Against intervals 0.46–0.81 wide these are immaterial. The zero-catch observation of Section 4.4 is safer still: the largest probability among the 27 test onset hours is 0.332, a margin of 0.168 below the threshold. A confirmatory fp32 run remains desirable, but no conclusion here depends on it.

- A magnetogram-derived SHARP baseline was not run; the GOES-history model is a weaker information set, which strengthens rather than weakens the argument.

- Our persistence reference reads the complete hourly record rather than the split file. This keeps three validation hours that a split-local lookup would drop; the choice changes no reported score at three decimal places.

8 Reproducibility

All probabilities, labels, code, and outputs are released. `heliofloor_data.py` is the canonical loader, metrics and block bootstrap, imported by everything else; `heliofloor_colab.py` regenerates the probabilities on a GPU; `goes_baseline.py` runs the matched comparison; `full_split_baseline.py` the complete-split protocol; `analysis_pack.py` the scoreboards, calibration, transfer and regime split; `block_support.py` the independent-block support behind each claim; `onset_ci.py` the onset rates and rule-of-three correction; `paired_diff.py` the paired method differences; `make_figures.py` the four figures; and `verify_paper.py` recomputes every claim in this manuscript and prints a pass/fail line for each. The three `probs_*.csv` files carry the 1,146 scored hours.

Block plans regenerate identically from seed 42, and every bootstrap interval reseeds per call so it reproduces independently of execution order. The scored probabilities are committed, so all analysis reproduces on a CPU in minutes with no GPU and no re-inference. Code and data: `https://github.com/kadircanyildirm-crypto/heliofloor`.

Acknowledgements

Analysis code and manuscript drafting were assisted by an AI coding assistant. All results were produced by the released scripts, every reported number is recomputed by `verify_paper.py`, and the author verified the label-leakage check, the split definitions and the quoted comparison figures against primary sources.

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